

Topic: Orthogonality - Euclidean Inner Product

Question 1. Consider the vector space $\mathbb{R}^3$, with the Euclidean inner product, and a fixed real number θ .

- Verify that the set $S = \{(\cos \theta, -\sin \theta, 0), (\sin \theta, \cos \theta, 0), (0, 0, 1)\}$ is an orthonormal basis of $\mathbb{R}^3$.
- With $\theta = \pi/3$, find the coordinate vector $[(1, 2, 3)]_S$.

Question 2. Consider the space $\mathbb{R}^3$ with the Euclidean inner product. Consider the subspace $\mathcal{W} = \{(x, y, z) : y = 4x\}$

- Verify that $\mathcal{B} = \{(0, 0, 1), (1, 4, 0)\}$ is an orthogonal set.
- Normalize vectors in $\mathcal{B}$ to obtain an orthonormal basis $\widehat{\mathcal{B}}$ for $\mathcal{W}$.
- Let $\mathbf{v} = (2, 6, 1)$. Compute the orthogonal projection $\text{Proj}_{\mathcal{W}}(\mathbf{v})$ of $\mathbf{v}$ onto $\mathcal{W}$. What is the distance of $\mathbf{v}$ from the plane $\mathcal{W}$? In other words, what is $\min\{\|\mathbf{v} - \mathbf{w}\| : \mathbf{w} \in \mathcal{W}\}$?

Question 3. Consider $\mathbb{R}^3$ with the Euclidean inner product and the basis $\mathcal{B} = \{(1, 0, 1), (0, 1, 1), (1, 3, 3)\}$.

- Apply the Gram-Schmidt process to $\mathcal{B}$ to obtain an orthogonal basis for $\mathbb{R}^3$.
- Normalize the vectors in the basis in part(a) to obtain an orthonormal basis $\mathcal{A}$ for $\mathbb{R}^3$.
- Let $\mathbf{v} = (1, 1, 2)$. What is $[\mathbf{v}]_{\mathcal{A}}$?
- Let $\mathbf{M} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 3 \\ 1 & 1 & 3 \end{bmatrix}$, and $\mathbf{Q}$ be the matrix whose columns are basis vectors in the orthonormal basis $\mathcal{A}$. Find a matrix $\mathbf{R}$ such that $\mathbf{M} = \mathbf{QR}$.

Question 4. (omitted from AY2023/2024)

Consider the complex Euclidean space $\mathbb{C}^3$ and the set $S = \{(1, 1, 1), (0, 3 - i, 3 + i)\}$.

- Apply the Gram-Schmidt process to S to obtain an orthogonal basis for $\text{span}(S)$.
- Normalize the vectors in the basis in part(a) to obtain an orthonormal basis $\mathcal{B}$ for $\text{span}(S)$.
- Set $\mathbf{w} = (20, 41 - 7i, 41 + 7i)$. Verify that $\mathbf{w} \in \text{span}(S)$ and proceed to find the coordinate vector of $\mathbf{w}$ with respect to the orthonormal basis $\mathcal{B}$.

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Question 5. Provide reason(s) why each of the following product function is *not* a inner product on the given real vector space.

(a) On $\mathbb{R}^2$: $\langle (a, b), (c, d) \rangle = ac - bd$.

(b) On $\mathcal{P}(\mathbb{R})$: $\langle f(x), g(x) \rangle = \int_0^1 f'(t)g(t) dt$.

(c) On $\mathcal{C}_{\mathbb{R}}[0, 1]$: $\langle f(x), g(x) \rangle = \int_0^{1/2} f(x)g(x) dx$.

(Here, $\mathcal{C}_{\mathbb{R}}[0, 1]$ is the real vector space of continuous functions with domain $[0, 1]$. Take note of the limits of the integration.)

Question 6. On the real vector space $\mathcal{P}_1(\mathbb{R})$, determine whether each of the following candidates for an inner product is indeed an inner product. Justify your answer.

(a) $\langle p(x), q(x) \rangle = p(0)q(0)$.

(b) $\langle p(x), q(x) \rangle = p(0)q(0) + p(1)q(1)$.

(c) $\langle p(x), q(x) \rangle = \int_{-1}^1 p(x)^2 q(x) dx$.

Question 7. Revisit Tutorial 09! Look at Questions 1–6 (except Q2(a)), replace **Euclidean inner product** with a **general real inner product** and show that the statements are true.

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Question 8. Consider $C[0, \pi]$, with the inner product $\langle f, g \rangle = \int_0^{\pi} f(x)g(x) dx$.

Let $f_1(x) = \frac{1}{\pi}$, $f_2(x) = \pi \cos x$ and $f_3(x) = \sin x$.

(a) Evaluate $\|f_i\|$ for each $i = 1, 2, 3$.

(b) Which two functions are orthogonal to each other?

Question 9. Consider the real inner product space $\mathcal{V} = \mathcal{C}_{\mathbb{R}}[0, 1]$, with the inner product $\langle f, g \rangle = \int_0^1 f(x)g(x) dx$. Let $\mathcal{W} = \text{span}(1, 2x - 1)$.

(a) Verify that the set $\mathcal{B} = \{1, 2x - 1\}$ is orthogonal. Proceed to normalize vectors in $\mathcal{B}$ to obtain an orthonormal basis $\hat{\mathcal{B}}$ for $\mathcal{W}$.

(b) Let $h(x) = x^2$. Compute the orthogonal projection $\text{Proj}_{\mathcal{W}} h(x)$, and determine

$$\min \left\{ \|h(x) - w\| : w \in \mathcal{W} \right\}.$$

Question 10. Consider $\mathcal{P}_2(\mathbb{R})$ with the evaluation inner product

$$\langle f(x), g(x) \rangle = f(-1)g(-1) + f(0)g(0) + f(1)g(1).$$

Let $p(x) = x - x^2$ and set $\mathcal{U} = \text{span}(p(x))$.

- (a) Evaluate $\|p(x)\|$, and find the unit vector $q(x)$.
- (b) We define the orthogonal complement of $\mathcal{U}$ to be $\mathcal{U}^\perp = \{g(x) \in \mathcal{P}_2(\mathbb{R}) : \langle p(x), g(x) \rangle = 0\}$. The following parts of Q6(b) will use the Gram-Schmidt procedure to find a basis for $\mathcal{U}^\perp$.
 - (i) We extend list $[x - x^2]$ to obtain the list $\mathcal{A} = [x - x^2, 1, x]$. Explain why $\mathcal{A}$ is a basis for $\mathcal{P}_2(\mathbb{R})$.
 - (ii) Apply the Gram-Schmidt procedure to find an orthogonal basis $\mathcal{B}$ for $\mathcal{P}_2(\mathbb{R})$.
 - (iii) Since $\mathcal{U}$ has dimension one, we have that $\mathcal{U}^\perp$ has dimension $3 - 1 = 2$ (proof is omitted from course). Furthermore, $\mathcal{U}$ is spanned by the first polynomial in $\mathcal{A}$. Therefore, removing the first polynomial in the orthogonal basis $\mathcal{B}$ results in a basis for $\mathcal{U}^\perp$. Using these facts, find a basis $\mathcal{B}$ for $\mathcal{U}^\perp$.
 - (iv) Find the orthogonal projection of $f(x) = x^2$ on $\mathcal{U}$.

Question 11. Let $\mathcal{C}_{\mathbb{R}}[-1, 1]$ be an inner product space with the inner product

$$\langle f, g \rangle = \int_{-1}^1 f(t)g(t)dt.$$

Let $\mathcal{W}_e$ and $\mathcal{W}_o$ denote the set of all continuous even and odd functions on $[-1, 1]$, respectively.

- (a) Prove that $\mathcal{W}_e$ is a subspace of $\mathcal{C}_{\mathbb{R}}[-1, 1]$. You may assume that $\mathcal{C}_{\mathbb{R}}[-1, 1]$ is a vector space.
- (b) Prove that every function $f \in \mathcal{C}_{\mathbb{R}}[-1, 1]$ can always be written as a sum of a continuous even function and a continuous odd function.
- (c) Prove that $\mathcal{W}_e^\perp = \mathcal{W}_o$, where $\mathcal{W}_o$ is the set of all continuous odd functions on $[-1, 1]$.